

Challenges Faced

- **Dataset Availability:** The original NCDC weather dataset was not readily available, so an NCDC-style sample dataset was created for the practical.
- **Hadoop Configuration:** Initial Hadoop setup and configuration required checking Hadoop version, HDFS directories, and ensuring that the Hadoop services were running properly.
- **File Path Issues:** Initially, an incorrect local path (/home/tushar/ncdc_project/) was used for the Mapper and Reducer files. The files were actually located directly in /home/tushar.
- **Hadoop Streaming Command:** The Hadoop Streaming command initially generated an error because the -files option was placed incorrectly. The command was corrected and the MapReduce job was executed successfully.
- **Output Directory Conflict:** Hadoop does not allow a MapReduce job to write to an existing output directory. The previous output directory had to be removed before running the job again.
- **Mapper and Reducer Testing:** The Mapper and Reducer had to be tested separately using Linux pipes and sort to verify that the key-value pairs and temperature calculations were correct.
- **Different Calculations:** Separate Reducer logic was required for finding the **maximum, minimum, and average temperatures**, as well as the highest temperature overall.